

Cointegration Statistical Arbitrage on US Equities: A Null Result, a Real Edge, and the Statistics that Tell Them Apart

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Abstract

This report builds a reproducible pipeline for cointegration-based pairs trading and applies it to two US-equity universes over 2013–2024. In the semiconductor sector, no pair is cointegrated once the screen is corrected for multiple testing, and the best of 210 searched strategies deflates to a Sharpe of **0.21**, indistinguishable from luck. On a small set of economically motivated pairs chosen in advance, the same pipeline finds a genuine edge: a Mastercard/Visa (MA/V) spread returns a Sharpe of **1.26** (+11.4% per year, growing \$100k to \$327k) with a probabilistic Sharpe ratio of **1.00** and a deflated Sharpe of **0.97**. The point of this exercise is false discovery rate screening and walk forward out of sample backtesting.

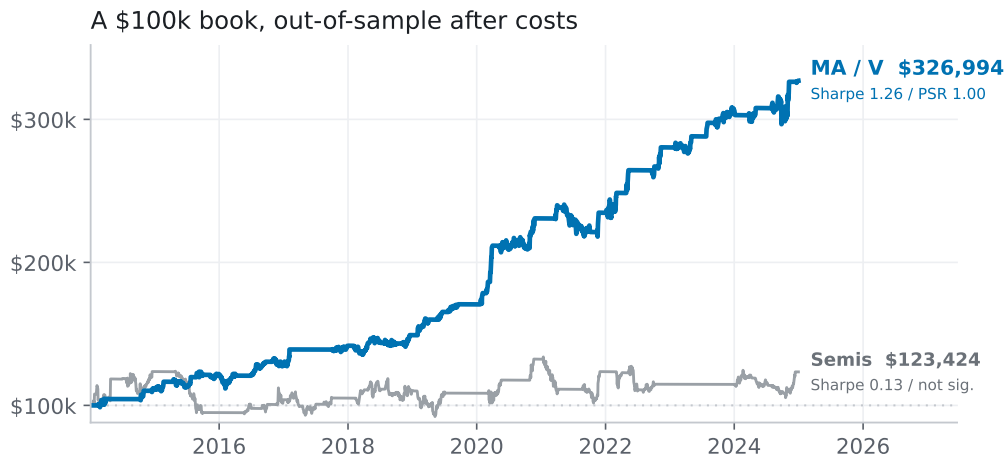


Figure 1: A \$100k capital traded out-of-sample after costs. Both equity curves rise, but only MA/V survives significance testing.

1 Introduction

Pairs trading rests on a simple idea. When two companies are economically tied, their share prices tend to move together, and the gap between them, the spread, tends to return to a stable level. A trader can sell the spread when it widens and buy it back when it closes, in principle earning a return that does not depend on the market’s overall direction.

The difficulty is that most apparent edges of this kind do not survive contact with reality. A backtest can look profitable because it quietly used information that would not have been available at the time, because it ignored the cost of trading, or simply because enough pairs were tried that one was bound to look good by chance. This report asks a deliberately plain question: do cointegration pairs trades survive realistic costs and rigorous significance testing? To answer it I built a pipeline that screens for cointegration with a multiple-testing correction, trades the spread

walk-forward with costs, and judges every result with tests that account for both sampling noise and the number of strategies searched.

The answer depends on where one looks. In this study, in the semiconductor sector nothing survives, and the result is a clean null. On a short list of economically motivated pairs chosen beforehand, one strategy, long/short Mastercard against Visa, is genuinely profitable and statistically significant. Section 2 develops the method, Section 3 describes the data, and Sections 4–6 present the two results and what separates them.

2 Methodology

The pipeline is built from standard components. This section states each one briefly and gives the reasoning behind it.

2.1 Cointegration and the Engle–Granger test

A price series behaves much like a random walk: it has a unit root and wanders without returning to any particular level. A tradeable spread has to be the opposite, stationary, pulled back toward its mean. Engle–Granger tests for this in two steps [1]: regress one log price on the other to obtain a hedge ratio β , then apply the Augmented Dickey–Fuller test to the residual $s_t = y_t - (\alpha + \beta x_t)$ (the null is a unit root, so a small p -value means the spread is stationary and the pair is cointegrated).

Correlation and cointegration are easy to confuse, but they are not the same. Correlation describes returns moving together day to day; cointegration is a property of the price *levels*, that some fixed combination of them stays stationary. A pairs trade needs the second, because only a stationary spread has a mean to revert to. High correlation without cointegration is common, and, as Section 4 shows, actively misleading.

2.2 Multiple testing and the false discovery rate

Screening many pairs creates a subtler problem. Testing $m = \binom{n}{2}$ pairs at the 5% level produces about $0.05m$ false positives even when nothing is truly cointegrated, so the raw count of “significant” pairs says very little. The Benjamini–Hochberg procedure rejects the ordered p -values up to the largest k with $p_{(k)} \leq \frac{k}{m}\alpha$, where α is the desired false discovery rate (FDR) significance level, which controls the expected *fraction* of false discoveries among the pairs it flags, in effect scaling the bar by how many tests were run [2].

2.3 Signal: half-life, z-score, and bands

Given a cointegrated pair, the spread’s speed of mean reversion can be calculated from an AR(1)/Ornstein–Uhlenbeck fit $\Delta s_t = a + b s_{t-1} + \varepsilon_t$, with half-life $-\ln 2/b$; this sets the horizon the strategy trades on. Positions are driven by the z-score $z_t = (s_t - \mu_s)/\sigma_s$: enter when the spread is stretched beyond an entry band and exit as it returns toward the mean, with a gap between the two bands so the position does not flip on small fluctuations.

2.4 Backtest: no look-ahead and costs

The backtest is built to avoid the two ways a track record lies. PnL uses the *lagged* position, $\text{gross}_t = \text{pos}_{t-1}(s_t - s_{t-1})$, so that a position chosen at today’s close earns the next day’s move rather than today’s, letting today’s position earn today’s move would be look-ahead. Every change

in position pays a cost proportional to turnover $|\Delta\text{pos}_t|$. Evaluation is walk-forward: β and the spread's mean and standard deviation are estimated on a training window, the next block is traded out-of-sample with those fixed parameters, and the window rolls forward. Because each trade uses only earlier data, the concatenated result is a genuine out-of-sample record.

2.5 Significance: the probabilistic and deflated Sharpe ratios

Two effects can make a Sharpe ratio misleading. The first is ordinary sampling error: a Sharpe estimated from a few years of noisy, non-normal returns carries a wide margin of error, which the probabilistic Sharpe ratio makes explicit [3],

$$\text{PSR} = \Phi \left(\frac{(\widehat{\text{SR}} - \text{SR}^*)\sqrt{n-1}}{\sqrt{1 - \gamma_3\widehat{\text{SR}} + \frac{\gamma_4-1}{4}\widehat{\text{SR}}^2}} \right),$$

the probability that the true Sharpe exceeds a benchmark SR^* , given sample size, skew, and kurtosis. The second effect is selection: the best of many strategies is biased upward even with no skill. The deflated Sharpe corrects for this by setting SR^* to the expected maximum of N skill-less trials, so a strategy is only credited to the extent it beats what a search of that size would produce by chance.

2.6 Dynamic hedge: the Kalman filter

The static hedge fits a single β . Because real relationships drift, the pipeline also offers a time-varying hedge: treating (α_t, β_t) as a random-walk state observed through $y_t = \alpha_t + \beta_t x_t + v_t$, a Kalman filter estimates the hedge online, and its one-step forecast error serves as a look-ahead-free spread. The added complexity is worth it only when the relationship actually moves; on a stable pair a dynamic hedge tends to chase noise and absorb the signal, a point Section 6 returns to.

3 Data

The study uses split- and dividend-adjusted daily closes from 2013 to 2024, roughly 3,000 trading days. Two universes are examined. The first is the semiconductor sector: twenty large-cap names plus the SMH and SOXX ETFs, a group chosen because its members share demand drivers and so are a natural place to look for cointegration. The second is a set of fourteen names forming seven pre-declared economic pairs (MA/V, KO/PEP, XOM/CVX, HD/LOW, GS/MS, WMT/TGT, GLD/IAU), each pairing two companies in the same business. These were fixed in advance on economic grounds; they were not selected by mining the screen for whatever looked most significant, which matters for the significance analysis in Section 5.

4 Results A, Semiconductors: a null

Of the 210 semiconductor pairs, none survive the false-discovery-rate correction. Ten clear the raw 5% bar, almost exactly the ≈ 10 that chance alone predicts, and all of them fall away once the correction is applied. Trading the tightest pair, AMAT/LRCX, out-of-sample earns a Sharpe of only 0.13 (probabilistic Sharpe 0.67), and searching all 210 pairs and keeping the best gives a Sharpe of 0.54 against a no-skill benchmark of 0.78, for a deflated Sharpe of **0.21**.

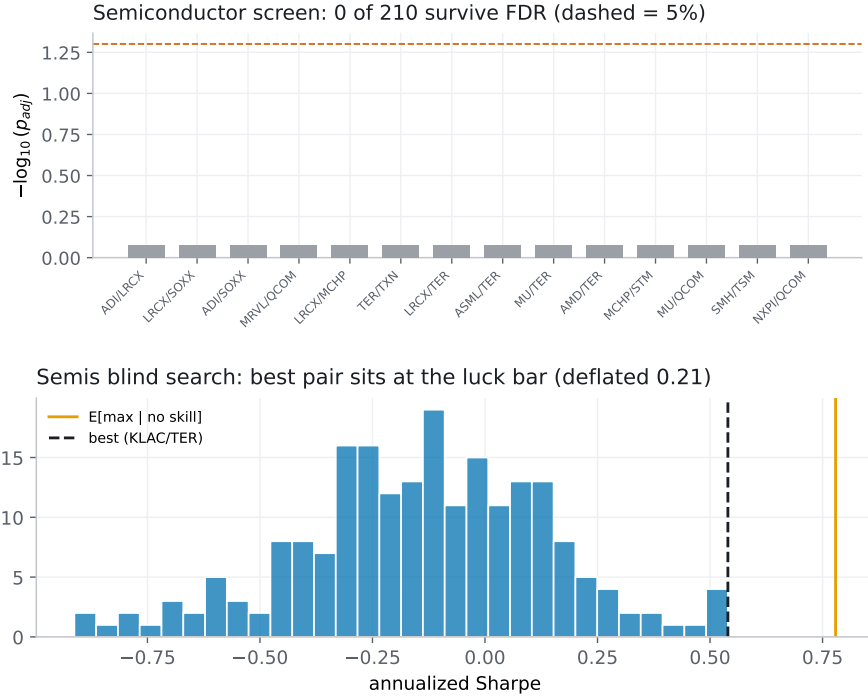


Figure 2: Top: no semiconductor pair clears the FDR threshold. Bottom: the best of a blind 210-pair search sits at the luck bar and deflates to 0.21.

Over this period, and after the correction, the semiconductor sector offered no pairs-trading edge, and the pairs that looked most promising on a raw screen were no better than the best a random search would turn up.

5 Results B, Economic pairs: a real edge

Screening the fourteen-name universe, only two pairs survive the correction, and both pair Walmart with a gold ETF, combinations with no economic reason to move together, and a reminder that a screen's smallest p -values can be coincidental. The tradeable result comes instead from a pair chosen in advance. Traded walk-forward after costs, **MA/V** (Mastercard against Visa) earns a Sharpe of **1.26** (+11.4% per year; \$100k grows to \$327k), is profitable in all eleven years, and has a 95% bootstrap confidence interval of [0.78, 1.79] with a probabilistic Sharpe near 1.00. It stays significant across the entire cost-and-band grid, and deflating against the seven pre-declared pairs leaves a Sharpe of **0.97**.



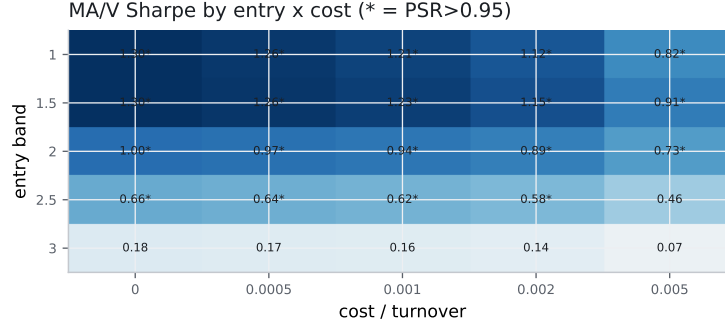


Figure 3: Top: MA/V grows \$100k to \$327k out-of-sample. Bottom: the Sharpe is significant across essentially the whole cost/entry grid.

	Semiconductors (blind search)	MA/V (pre-declared)
Best / pair Sharpe	0.54 (KLAC/TER)	1.26
Deflated Sharpe	0.21	0.97

Table 1: The same significance machinery, opposite verdicts.

The contrast with the semiconductor search is the central finding. There, the best of 210 pairs deflated to 0.21 precisely because it was the winner of a large search. MA/V was fixed beforehand and measured against only a short list, so it pays almost no selection penalty and its deflated Sharpe holds at 0.97. The dollar figure is best read as illustration: it assumes a fixed notional and frictionless compounding, and the case for the strategy rests on the Sharpe and its significance, not on the headline total.

6 Discussion

The two results point to the same lesson. The profitable, defensible strategy did not come from mining a large universe for the best-looking pair; it came from a single hypothesis formed in advance and then tested out of sample. The deflated Sharpe is what makes that distinction measurable, it is the gap between the semiconductor search at 0.21 and the pre-declared pair at 0.97.

The dynamic-hedge experiment makes a related point about matching a method to its problem. A Kalman filter that lets the hedge ratio drift helped on AMAT/LRCX, whose relationship re-rated over the decade, but underperformed a static hedge on MA/V, whose ratio was essentially constant; there the extra flexibility only added turnover and absorbed the signal. Choosing when to use a tool mattered more than the tool itself.

Several limitations bound these conclusions. The null rests on a single sector and the positive result on a short list of pre-declared pairs, so a broader study would test more of each. The data are daily and the cost model is a flat charge on turnover, so nothing here reflects intraday execution or market impact. The evaluation covers 2013–2024 only, and while MA/V survives every test applied here, its performance in the years since would be the real confirmation.

7 Conclusion

This project built an end-to-end, reproducible pipeline for cointegration-based pairs trading and tested it under realistic conditions. Pairs are screened for cointegration with a correction for multiple testing, the resulting spreads are traded out of sample with transaction costs, and each result is judged with significance tests that account for both sampling noise and the number of strategies searched. Applied across different equity universes, the study shows that whether a pairs trade holds up depends far less on the trading rule than on the testing framework: honest screening, no look-ahead, and a correction for selection bias.

References

- [1] Engle, R. F. and Granger, C. W. J. (1987). Co-integration and error correction: Representation, estimation, and testing. *Econometrica*, 55(2), 251–276.
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- [3] Bailey, D. H. and López de Prado, M. (2014). The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *The Journal of Portfolio Management*, 40(5), 94–107.